



PREMIUM

Full-Length Mock Test

Mock 03

Computer Science and Information Technology

GATE CS

Total Questions	60
Maximum Marks	108
Duration	180 minutes (3 Hours)
Exam Pattern	MCQ + MSQ + NAT
Negative Marking	MCQ/MSQ: -0.33 per wrong answer
Designed For	EduNeuro Premium

Easy 20%

Moderate 33%

Hard 47%

Based on Previous Year Question Pattern Analysis | Generated by EduNeuro Premium

General Instructions

This question paper contains THREE sections: Section A, Section B, and Section C.

Section A: Multiple Choice Questions (MCQ) — carries one or two marks each. Only ONE option is correct.

Section B: Multiple Select Questions (MSQ) — carries two marks each. ONE OR MORE than one option(s) is/are correct.

Section C: Numerical Answer Type (NAT) — carries one or two marks. Enter the numerical value in the space provided.

Negative Marking: -0.33 marks will be deducted for each wrong MCQ and each wrong MSQ answer.

No negative marking for NAT questions.

There are 60 questions for a total of 100 marks.

Duration: 3 hours (180 minutes).

Use the space provided for rough work. Scribbling on the question paper is permitted.

Do not ask for clarification from the invigilator. Interpret questions as given.

Paper Structure

Section	Type	Questions	Marks/Q	Total Marks	Negative
A	MCQ	35	1 / 2	35	-0.33
B	MSQ	5	2	10	-0.33
C	NAT	20	1 / 2	35	None
Total		60		100	

Marking Scheme

Correct MCQ: +1 mark (or +2 marks as specified per question)

Wrong MCQ: -0.33 marks

Correct MSQ: +2 marks

Wrong MSQ (partial or no match): -0.33 marks

Correct NAT: +1 mark (or +2 marks as specified per question)

No answer / Unattempted: 0 marks

NAT wrong answer: 0 marks (no negative marking)

Section A — Multiple Choice Questions (MCQ)

Q.1 [1 Mark] [EASY] [Algorithms — Greedy]

Which data structure implements FIFO?

- A) Stack
 - B) Queue
 - C) Tree
 - D) Graph
-

Q.2 [1 Mark] [EASY] [Data Structures — Hashing]

Time complexity of binary search?

- A) $O(n)$
 - B) $O(\log n)$
 - C) $O(1)$
 - D) $O(n^2)$
-

Q.3 [1 Mark] [EASY] [DBMS — Normalization]

Normal form where all non-prime attributes depend on key:

- A) 1NF
 - B) 2NF
 - C) 3NF
 - D) BCNF
-

Q.4 [1 Mark] [EASY] [OS — Memory Management]

In deadlock prevention, hold-and-wait is prevented by:

- A) Preemption
 - B) Resource ordering
 - C) Mutual exclusion
 - D) Allocation at once
-

Q.5 [1 Mark] [EASY] [CN — HTTP/HTTPS]

Class A IP range:

- A) 0-127
 - B) 128-191
 - C) 192-223
 - D) 224-239
-

Q.6 [1 Mark] [EASY] [TOC — Grammars]

A regular grammar generates:

- A) Type 0
 - B) Type 1
 - C) Type 2
 - D) Type 3
-

Q.7 [1 Mark] [EASY] [COA — I/O]

Cache hit ratio if access time = 10ns, hit time = 1ns, miss penalty = 100ns:

A) 0.1

B) 0.5

C) 0.9

D) 0.99

Q.8 [1 Mark] [EASY] [Digital Logic — Sequential Circuits]

Which is not a page replacement algorithm?

- A) FIFO
 - B) LRU
 - C) Optimal
 - D) FCFS
-

Q.9 [1 Mark] [EASY] [Engineering Mathematics — Numerical Methods]

Which data structure implements FIFO?

- A) Stack
 - B) Queue
 - C) Tree
 - D) Graph
-

Q.10 [1 Mark] [EASY] [Programming — C Programming]

Time complexity of binary search?

- A) $O(n)$
 - B) $O(\log n)$
 - C) $O(1)$
 - D) $O(n^2)$
-

Q.11 [1 Mark] [EASY] [Algorithms — Time Complexity]

Normal form where all non-prime attributes depend on key

- A) 1NF
 - B) 2NF
 - C) 3NF
 - D) BCNF
-

Q.12 [1 Mark] [EASY] [Data Structures — Graphs]

In deadlock prevention, hold-and-wait is prevented by:

- A) Preemption
 - B) Resource ordering
 - C) Mutual exclusion
 - D) Allocation at once
-

Q.13 [2 Marks] [MODERATE] [DBMS — Indexing]

Class A IP range:

- A) 0-127
 - B) 128-191
 - C) 192-223
 - D) 224-239
-

Q.14 [2 Marks] [MODERATE] [OS — Deadlock]

A regular grammar generates:

- A) Type 0

B) Type 1

C) Type 2

D) Type 3

Q.15 [2 Marks] [MODERATE] [CN — HTTP/HTTPS]

Cache hit ratio if access time = 10ns, hit time = 1ns, miss penalty = 100ns:

- A) 0.1
 - B) 0.5
 - C) 0.9
 - D) 0.99
-

Q.16 [2 Marks] [MODERATE] [TOC — Decidability]

Which is not a page replacement algorithm?

- A) FIFO
 - B) LRU
 - C) Optimal
 - D) FCFS
-

Q.17 [2 Marks] [MODERATE] [COA — Instruction Formats]

Which data structure implements FIFO?

- A) Stack
 - B) Queue
 - C) Tree
 - D) Graph
-

Q.18 [2 Marks] [MODERATE] [Digital Logic — Minimization]

Time complexity of binary search?

- A) $O(n)$
 - B) $O(\log n)$
 - C) $O(1)$
 - D) $O(n^2)$
-

Q.19 [2 Marks] [MODERATE] [Engineering Mathematics — Numerical Methods]

Normal form where all non-prime attributes depend on key:

- A) 1NF
 - B) 2NF
 - C) 3NF
 - D) BCNF
-

Q.20 [2 Marks] [MODERATE] [Programming — C Programming]

In deadlock prevention, hold-and-wait is prevented by:

- A) Preemption
 - B) Resource ordering
 - C) Mutual exclusion
 - D) Allocation at once
-

Q.21 [2 Marks] [MODERATE] [Algorithms — Graph Algorithms]

Class A IP range:

- A) 0-127

B) 128-191

C) 192-223

D) 224-239

Q.22 [2 Marks] [MODERATE] [Data Structures — Linked Lists]

A regular grammar generates:

- A) Type 0
 - B) Type 1
 - C) Type 2
 - D) Type 3
-

Q.23 [2 Marks] [MODERATE] [DBMS — Concurrency]

Cache hit ratio if access time = 10ns, hit time = 1ns, miss penalty = 100ns:

- A) 0.1
 - B) 0.5
 - C) 0.9
 - D) 0.99
-

Q.24 [2 Marks] [MODERATE] [OS — CPU Scheduling]

Which is not a page replacement algorithm?

- A) FIFO
 - B) LRU
 - C) Optimal
 - D) FCFS
-

Q.25 [2 Marks] [MODERATE] [CN — Network Security]

Which data structure implements FIFO?

- A) Stack
 - B) Queue
 - C) Tree
 - D) Graph
-

Q.26 [2 Marks] [MODERATE] [TOC — Regular Languages]

Time complexity of binary search?

- A) $O(n)$
 - B) $O(\log n)$
 - C) $O(1)$
 - D) $O(n^2)$
-

Q.27 [2 Marks] [MODERATE] [COA — Addressing Modes]

Normal form where all non-prime attributes depend on key:

- A) 1NF
 - B) 2NF
 - C) 3NF
 - D) BCNF
-

Q.28 [2 Marks] [MODERATE] [Digital Logic — Minimization]

In deadlock prevention, hold-and-wait is prevented by:

- A) Preemption

B) Resource ordering

C) Mutual exclusion

D) Allocation at once

Q.29 [2 Marks] [MODERATE] [Engineering Mathematics — Probability]

Class A IP range:

- A) 0-127
 - B) 128-191
 - C) 192-223
 - D) 224-239
-

Q.30 [2 Marks] [MODERATE] [Programming — Data Structures]

A regular grammar generates:

- A) Type 0
 - B) Type 1
 - C) Type 2
 - D) Type 3
-

Q.31 [2 Marks] [MODERATE] [Algorithms — Divide & Conquer]

Cache hit ratio if access time = 10ns, hit time = 1ns, miss penalty = 100ns:

- A) 0.1
 - B) 0.5
 - C) 0.9
 - D) 0.99
-

Q.32 [2 Marks] [MODERATE] [Data Structures — Graphs]

Which is not a page replacement algorithm?

- A) FIFO
 - B) LRU
 - C) Optimal
 - D) FCFS
-

Q.33 [2 Marks] [HARD] [DBMS — Transactions]

Which data structure implements FIFO?

- A) Stack
 - B) Queue
 - C) Tree
 - D) Graph
-

Q.34 [2 Marks] [HARD] [OS — Memory Management]

Time complexity of binary search?

- A) $O(n)$
 - B) $O(\log n)$
 - C) $O(1)$
 - D) $O(n^2)$
-

Q.35 [2 Marks] [HARD] [CN — Routing]

Normal form where all non-prime attributes depend on key:

- A) 1NF

B) 2NF

C) 3NF

D) BCNF

EduNeuro

Section B — Multiple Select Questions (MSQ)

Q.36 [2 Marks] [HARD] [TOC — Regular Languages]

Which are NP-Complete problems?

- A) Hamiltonian Cycle
 - B) Sorting
 - C) Clique
 - D) Binary Search
 - E) Boolean SAT
-

Q.37 [2 Marks] [HARD] [COA — Instruction Formats]

Which are stable sorting algorithms?

- A) Bubble
 - B) Insertion
 - C) Merge
 - D) Quick
 - E) Selection
-

Q.38 [2 Marks] [HARD] [Digital Logic — Sequential Circuits]

Which are valid lossless join decompositions?

- A) BCNF
 - B) 3NF
 - C) 2NF
 - D) 1NF
 - E) All always lossless
-

Q.39 [2 Marks] [HARD] [Engineering Mathematics — Discrete Math]

TCP provides:

- A) Connection-oriented
 - B) Reliable
 - C) Unreliable
 - D) Flow control
 - E) Error recovery
-

Q.40 [2 Marks] [HARD] [Programming — C Programming]

Which are NP-Complete problems?

- A) Hamiltonian Cycle
 - B) Sorting
 - C) Clique
 - D) Binary Search
 - E) Boolean SAT
-

Section C — Numerical Answer Type (NAT)**Q.51 [2 Marks] [HARD] [Algorithms — Dynamic Programming]**

Array A[1..100] traversed once. Time complexity?

Answer: _____

Q.52 [2 Marks] [HARD] [Data Structures — Linked Lists]

BST with 7 nodes at height 2 (root=height 0). Min nodes?

Answer: _____

Q.53 [2 Marks] [HARD] [DBMS — Normalization]

GATE CSE 2023: Semaphore with S=3, 5 processes need 1 each. How many wait?

Answer: _____

Q.54 [2 Marks] [HARD] [OS — CPU Scheduling]

IPv4 header length = 20 bytes without options. In 32-bit words?

Answer: _____

Q.55 [2 Marks] [HARD] [CN — HTTP/HTTPS]

Quick sort: partition on 10 elements. Pivot comparisons ~?

Answer: _____

Q.56 [2 Marks] [HARD] [TOC — Grammars]

Array A[1..100] traversed once. Time complexity?

Answer: _____

Q.57 [2 Marks] [HARD] [COA — Pipelining]

BST with 7 nodes at height 2 (root=height 0). Min nodes?

Answer: _____

Q.58 [2 Marks] [HARD] [Digital Logic — Boolean Algebra]

GATE CSE 2023: Semaphore with S=3, 5 processes need 1 each. How many wait?

Answer: _____

Q.59 [2 Marks] [HARD] [Engineering Mathematics — Calculus]

IPv4 header length = 20 bytes without options. In 32-bit words?

Answer: _____

Q.60 [2 Marks] [HARD] [Programming — C Programming]

Quick sort: partition on 10 elements. Pivot comparisons ~?

Answer: _____

EduNeuro

Answer Key

Verify your answers after completing the full paper.

Q.1:	B	Q.2:	B	Q.3:	B
Q.4:	D	Q.5:	A	Q.6:	D
Q.7:	C	Q.8:	D	Q.9:	B
Q.10:	B	Q.11:	B	Q.12:	D
Q.13:	A	Q.14:	D	Q.15:	C
Q.16:	D	Q.17:	B	Q.18:	B
Q.19:	B	Q.20:	D	Q.21:	A
Q.22:	D	Q.23:	C	Q.24:	D
Q.25:	B	Q.26:	B	Q.27:	B
Q.28:	D	Q.29:	A	Q.30:	D
Q.31:	C	Q.32:	D	Q.33:	B
Q.34:	B	Q.35:	B	Q.36:	ACE
Q.37:	ABC	Q.38:	ABC	Q.39:	ABDE
Q.40:	ACE	Q.41:	ABC	Q.42:	ABC
Q.43:	ABDE	Q.44:	ACE	Q.45:	ABC
Q.46:	ABC	Q.47:	ABDE	Q.48:	ACE
Q.49:	ABC	Q.50:	ABC	Q.51:	100
Q.52:	7	Q.53:	2	Q.54:	5
Q.55:	9	Q.56:	100	Q.57:	
Q.58:	2	Q.59:	5	Q.60:	9

Detailed Solutions

Question 1 [Algorithms — Greedy]

Which data structure implements FIFO?

Answer: (B)

Queue follows First-In-First-Out (FIFO) principle.

Question 2 [Data Structures — Hashing]

Time complexity of binary search?

Answer: (B)

Binary search halves search space each step !' $O(\log n)$.

Question 3 [DBMS — Normalization]

Normal form where all non-prime attributes depend on key:

Answer: (B)

2NF: no partial dependency of non-prime attributes on key.

Question 4 [OS — Memory Management]

In deadlock prevention, hold-and-wait is prevented by:

Answer: (D)

Allocation at once (request all resources at beginning) prevents hold-and-wait.

Question 5 [CN — HTTP/HTTPS]

Class A IP range:

Answer: (A)

Class A: 0.0.0.0 - 127.255.255.255. First octet 0-127.

Question 6 [TOC — Grammars]

A regular grammar generates:

Answer: (D)

Regular grammars (Type 3) generate regular languages.

Question 7 [COA — I/O]

Cache hit ratio if access time = 10ns, hit time = 1ns, miss penalty = 100ns:

Answer: (C)

AMAT = Hit time + Miss rate $\times$ Miss penalty = 1 + (1-h) $\times$ 100 = 10 !' h = 0.9.

Question 8 [Digital Logic — Sequential Circuits]

Which is not a page replacement algorithm?

Answer: (D)

FCFS is CPU scheduling. FIFO, LRU, Optimal are page replacement.

Question 9 [Engineering Mathematics — Numerical Methods]

Which data structure implements FIFO?

Answer:

(B)

Queue follows First-In-First-Out (FIFO) principle.

Question 10 [Programming — C Programming]

Time complexity of binary search?

Answer: (B)

Binary search halves search space each step !' $O(\log n)$.

Question 11 [Algorithms — Time Complexity]

Normal form where all non-prime attributes depend on key:

Answer: (B)

2NF: no partial dependency of non-prime attributes on key.

Question 12 [Data Structures — Graphs]

In deadlock prevention, hold-and-wait is prevented by:

Answer: (D)

Allocation at once (request all resources at beginning) prevents hold-and-wait.

Question 13 [DBMS — Indexing]

Class A IP range:

Answer: (A)

Class A: 0.0.0.0 - 127.255.255.255. First octet 0-127.

Question 14 [OS — Deadlock]

A regular grammar generates:

Answer: (D)

Regular grammars (Type 3) generate regular languages.

Question 15 [CN — HTTP/HTTPS]

Cache hit ratio if access time = 10ns, hit time = 1ns, miss penalty = 100ns:

Answer: (C)

$AMAT = \text{Hit time} + \text{Miss rate} \times \text{Miss penalty} = 1 + (1-h) \times 100 = 10$!' $h = 0.9$.

Question 16 [TOC — Decidability]

Which is not a page replacement algorithm?

Answer: (D)

FCFS is CPU scheduling. FIFO, LRU, Optimal are page replacement.

Question 17 [COA — Instruction Formats]

Which data structure implements FIFO?

Answer: (B)

Queue follows First-In-First-Out (FIFO) principle.

Question 18 [Digital Logic — Minimization]

Time complexity of binary search?

Answer:

(B)

Binary search halves search space each step !' $O(\log n)$.

Question 19 [Engineering Mathematics — Numerical Methods]

Normal form where all non-prime attributes depend on key:

Answer: (B)

2NF: no partial dependency of non-prime attributes on key.

Question 20 [Programming — C Programming]

In deadlock prevention, hold-and-wait is prevented by:

Answer: (D)

Allocation at once (request all resources at beginning) prevents hold-and-wait.

Question 21 [Algorithms — Graph Algorithms]

Class A IP range:

Answer: (A)

Class A: 0.0.0.0 - 127.255.255.255. First octet 0-127.

Question 22 [Data Structures — Linked Lists]

A regular grammar generates:

Answer: (D)

Regular grammars (Type 3) generate regular languages.

Question 23 [DBMS — Concurrency]

Cache hit ratio if access time = 10ns, hit time = 1ns, miss penalty = 100ns:

Answer: (C)

AMAT = Hit time + Miss rate $\times$ Miss penalty = 1 + (1-h) $\times$ 100 = 10 !' h = 0.9.

Question 24 [OS — CPU Scheduling]

Which is not a page replacement algorithm?

Answer: (D)

FCFS is CPU scheduling. FIFO, LRU, Optimal are page replacement.

Question 25 [CN — Network Security]

Which data structure implements FIFO?

Answer: (B)

Queue follows First-In-First-Out (FIFO) principle.

Question 26 [TOC — Regular Languages]

Time complexity of binary search?

Answer: (B)

Binary search halves search space each step !' $O(\log n)$.

Question 27 [COA — Addressing Modes]

Normal form where all non-prime attributes depend on key:

Answer:

(B)

2NF: no partial dependency of non-prime attributes on key.

Question 28 [Digital Logic — Minimization]

In deadlock prevention, hold-and-wait is prevented by:

Answer: (D)

Allocation at once (request all resources at beginning) prevents hold-and-wait.

Question 29 [Engineering Mathematics — Probability]

Class A IP range:

Answer: (A)

Class A: 0.0.0.0 - 127.255.255.255. First octet 0-127.

Question 30 [Programming — Data Structures]

A regular grammar generates:

Answer: (D)

Regular grammars (Type 3) generate regular languages.

Question 31 [Algorithms — Divide & Conquer]

Cache hit ratio if access time = 10ns, hit time = 1ns, miss penalty = 100ns:

Answer: (C)

AMAT = Hit time + Miss rate $\times$ Miss penalty = $1 + (1-h) \times 100 = 10$!' h = 0.9.

Question 32 [Data Structures — Graphs]

Which is not a page replacement algorithm?

Answer: (D)

FCFS is CPU scheduling. FIFO, LRU, Optimal are page replacement.

Question 33 [DBMS — Transactions]

Which data structure implements FIFO?

Answer: (B)

Queue follows First-In-First-Out (FIFO) principle.

Question 34 [OS — Memory Management]

Time complexity of binary search?

Answer: (B)

Binary search halves search space each step !' $O(\log n)$.

Question 35 [CN — Routing]

Normal form where all non-prime attributes depend on key:

Answer: (B)

2NF: no partial dependency of non-prime attributes on key.

Question 36 [TOC — Regular Languages]

Which are NP-Complete problems?

Answer:

(A, C, E)

Hamiltonian Cycle, Clique, SAT are NP-Complete. Sorting, Binary Search are P.

Question 37 [COA — Instruction Formats]

Which are stable sorting algorithms?

Answer: (A, B, C)

Bubble, Insertion, Merge are stable. Quick, Selection are not.

Question 38 [Digital Logic — Sequential Circuits]

Which are valid lossless join decompositions?

Answer: (A, B, C)

Decomposition to BCNF may not preserve dependencies. 3NF is always lossless.

Question 39 [Engineering Mathematics — Discrete Math]

TCP provides:

Answer: (A, B, D, E)

TCP is connection-oriented, reliable, has flow control, error recovery.

Question 40 [Programming — C Programming]

Which are NP-Complete problems?

Answer: (A, C, E)

Hamiltonian Cycle, Clique, SAT are NP-Complete. Sorting, Binary Search are P.

Question 41 [Algorithms — Dynamic Programming]

Which are stable sorting algorithms?

Answer: (A, B, C)

Bubble, Insertion, Merge are stable. Quick, Selection are not.

Question 42 [Data Structures — Linked Lists]

Which are valid lossless join decompositions?

Answer: (A, B, C)

Decomposition to BCNF may not preserve dependencies. 3NF is always lossless.

Question 43 [DBMS — SQL]

TCP provides:

Answer: (A, B, D, E)

TCP is connection-oriented, reliable, has flow control, error recovery.

Question 44 [OS — File Systems]

Which are NP-Complete problems?

Answer: (A, C, E)

Hamiltonian Cycle, Clique, SAT are NP-Complete. Sorting, Binary Search are P.

Question 45 [CN — TCP/IP]

Which are stable sorting algorithms?

Answer:

(A, B, C)

Bubble, Insertion, Merge are stable. Quick, Selection are not.

Question 46 [TOC — Grammars]

Which are valid lossless join decompositions?

Answer: (A, B, C)

Decomposition to BCNF may not preserve dependencies. 3NF is always lossless.

Question 47 [COA — Addressing Modes]

TCP provides:

Answer: (A, B, D, E)

TCP is connection-oriented, reliable, has flow control, error recovery.

Question 48 [Digital Logic — Minimization]

Which are NP-Complete problems?

Answer: (A, C, E)

Hamiltonian Cycle, Clique, SAT are NP-Complete. Sorting, Binary Search are P.

Question 49 [Engineering Mathematics — Calculus]

Which are stable sorting algorithms?

Answer: (A, B, C)

Bubble, Insertion, Merge are stable. Quick, Selection are not.

Question 50 [Programming — Pointers]

Which are valid lossless join decompositions?

Answer: (A, B, C)

Decomposition to BCNF may not preserve dependencies. 3NF is always lossless.

Question 51 [Algorithms — Dynamic Programming]

Array $A[1..100]$ traversed once. Time complexity?

Answer: (100)

Traversing n elements !' $O(n) = 100$ operations.

Question 52 [Data Structures — Linked Lists]

BST with 7 nodes at height 2 (root=height 0). Min nodes?

Answer: (7)

Full binary tree at height 2 has 7 nodes (perfect binary tree).

Question 53 [DBMS — Normalization]

GATE CSE 2023: Semaphore with $S=3$, 5 processes need 1 each. How many wait?

Answer: (2)

3 proceed immediately. 2 wait. Total blocked = 2.

Question 54 [OS — CPU Scheduling]

IPv4 header length = 20 bytes without options. In 32-bit words?

Answer:

20 bytes / 4 bytes per word = 5 words.

Question 55 [CN — HTTP/HTTPS]

Quick sort: partition on 10 elements. Pivot comparisons ~?

Answer: (9)

Partitioning n elements requires $n-1$ comparisons. For 10: 9.

Question 56 [TOC — Grammars]

Array $A[1..100]$ traversed once. Time complexity?

Answer: (100)

Traversing n elements !' $O(n) = 100$ operations.

Question 57 [COA — Pipelining]

BST with 7 nodes at height 2 (root=height 0). Min nodes?

Answer: (7)

Full binary tree at height 2 has 7 nodes (perfect binary tree).

Question 58 [Digital Logic — Boolean Algebra]

GATE CSE 2023: Semaphore with $S=3$, 5 processes need 1 each. How many wait?

Answer: (2)

3 proceed immediately. 2 wait. Total blocked = 2.

Question 59 [Engineering Mathematics — Calculus]

IPv4 header length = 20 bytes without options. In 32-bit words?

Answer: (5)

20 bytes / 4 bytes per word = 5 words.

Question 60 [Programming — C Programming]

Quick sort: partition on 10 elements. Pivot comparisons ~?

Answer: (9)

Partitioning n elements requires $n-1$ comparisons. For 10: 9.
